

Muhammad Raihan Thaffan Hidayat

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Profile

Telecommunications Engineering student at Telkom University with experience across deep learning, computer vision, web development, and DevOps. Built a speech-based depression detection model during an internship at HUMIC Engineering, an edge people-counting system on Raspberry Pi 5, a full-stack finance web app, and a CI/CD pipeline on a self-managed VPS. Also a teaching assistant at MBC Laboratory

Work Experience

Project Management Officer, Internship **Jun 2025 - Sept 2025**
PT LEN Industri (Persero), Bandung

- Developed and managed project dashboards using Google Looker Studio for dynamic, real-time project monitoring with a focus on budget oversight.
- Utilized AppSheet and Looker to optimize project management processes during the internship period.
- Prepared various project documents required for management reporting.

Deep Learning Engineer, Internship **Apr 2026 - July 2026**
HUMIC (Human Centric) Engineering, Bandung

- Designed and implemented PyTorch Deep Learning architectures (Multi-Branch 1D/2D CNN & BiLSTM-MHA) for speech-based depression detection using the DAIC-WOZ dataset (189 participants) under strict participant-level 5-Fold Stratified Cross-Validation.
- Integrated self-supervised pre-trained representations (Wav2Vec 2.0 768-dim embeddings) with Mel-Spectrograms and 120-dim MFCCs into a multi-modal deep feature fusion head.
- Mitigated severe class imbalance (2.3:1 ratio) through custom AdamW optimization, ReduceLROnPlateau scheduling, 0.4 dropout regularization, and weighted Cross-Entropy Loss penalties.
- Engineered Explainable AI (XAI) attribution pipelines leveraging SHAP GradientExplainer directly on CNN fusion layers to interpret spectral-temporal audio feature attributions for clinical transparency.

Project

Portfolio Website CI/CD on Self-Managed VPS **2026**

- Built a CI/CD pipeline (GitHub, Jenkins, tests, SonarQube analysis, manual approval, deploy) for a Vite and Tailwind CSS portfolio site.
- Ran Jenkins and SonarQube with Docker Compose on an Ubuntu 24.04 VPS, and deployed the application as a Docker container.
- Hardened the server with key-only SSH, ufw firewall, and fail2ban.
- Set up Caddy as reverse proxy with automatic Let's Encrypt HTTPS certificates.

MyPundi - Personal Finance Web App | https://mypundi.my.id **Oct 2026**

- Built a personal finance tracker with multi-account support, a dashboard with budget limits and monthly cash-flow comparison, and deployed it at mypundi.my.id.
- Stack: Next.js (App Router), React, TypeScript, Tailwind CSS, Supabase with Row Level Security.
- Added TOTP two-factor authentication and a local screen lock using Web Crypto.
- Implemented receipt scanning with Gemini Vision API, with Tesseract.js in the browser as a fallback.
- Wrote tests with Vitest and Playwright; deployed through a Jenkins pipeline with a SonarQube quality gate on an Ubuntu VPS.

A Computer Vision-Based Human Monitoring and Counting System Using YOLOv8n and ByteTrack **Feb 2026 - Aug 2026**

- Developed an edge AI human detection and counting system deployed on a Raspberry Pi 5 without personal identity recognition for user privacy.
- Implemented YOLOv8n fine-tuned on a custom dataset and integrated ByteTrack with virtual line-crossing logic to prevent double-counting.
- Built an asynchronous dual-path distribution to Firebase Realtime Database and created the Detectra mobile app using Flutter for live monitoring.

ESP32 Based Smart Assistive Device for the Visually Impaired **Mar 2025 - Jun 2025**

- Designed a navigation aid for the visually impaired using an ESP32 microcontroller to enhance user safety and mobility.
- Integrated ultrasonic sensors for obstacle detection, a tilt sensor for balance monitoring, and a buzzer alarm system.
- Developed device logic using C++ and integrated it with the Blynk.io platform for monitoring functionality.

CryptoMania Website Development **Mar 2025 - Jun 2025**

- Built a cryptocurrency information platform providing real-time data on coin names, market values, descriptions, and price history.
- Led the front-end development using React.js, JavaScript, and CSS.
- Designed and implemented a responsive user interface to present data for the top 50 coins informatively.

Organizational Experience

Public Relations Division, PERMISI **2022 - 2023**

- Actively participated as a team member in the Education Roadshow project targeting various high schools in the Bekasi area.
- Carried out intensive coordination and direct socialization of the program to schools to build and maintain strategic cooperative relationships.
- Managed internal team communication to ensure that activity schedules and presentation materials were delivered effectively and on time.
- Achievements: Successfully reached and increased the enthusiasm of high school students in Bekasi in learning about Telkom University's education programs through measurable outreach activities

Public Relations Division, TELERIUM **2023 - 2024**

- Acting as a member who serves as the main communication bridge through intensive coordination between laboratories within the Faculty of Electrical Engineering.
- Ensuring the smooth flow of information between the organizers and laboratory assistants in order to minimize technical obstacles during the event.

Practicum Division, MBC LABORATORY **2025 - 2026**

- Responsible for teaching practical material directly to students in the laboratory to ensure in-depth technical understanding.
- Compiling and developing practical modules as the main technical guide in the student learning process.
- Installing and configuring the necessary software to support the smooth running of practical activities.
- Managing the physical maintenance of the laboratory, including workbench layout and cable management, to ensure that the practical area remains safe and conducive.

Education

Telkom University **2022 - 2026**
S1 Telecommunications Engineering

Soft Skill

- Time Management
- Discipline
- Adaptability
- Teamwork

Hard Skill

- Deep Learning and AI: PyTorch, CNN, BiLSTM, Wav2Vec 2.0, Explainable AI (SHAP)
- Computer Vision: YOLOv8, ByteTrack, edge deployment on Raspberry Pi 5
- Web and Mobile: React.js, Next.js, JavaScript, TypeScript, HTML, CSS, Tailwind CSS, Flutter
- Back-End and Data: Supabase, Firebase, SQL (MySQL)
- DevOps: Git, GitHub, Docker, Portainer, Jenkins, SonarQube, Prometheus, Grafana, Caddy, Ubuntu VPS hardening
- IoT: Arduino, ESP32, sensor integration (ultrasonic, buzzer, tilt)
- Other: Python, C/C++ (basic), Cisco Packet Tracer (basic), Microsoft Excel (basic), Google Looker Studio

Licenses & Certifications

Learn the Basics of AI, Dicoding Indonesia — **Nov 2024 - Nov 2027**

Getting Started with Python Programming, Dicoding Indonesia — **Nov 2024 - Nov 2027**

Learn the Basics of Data Science, Dicoding Indonesia — **Oct 2024 - Oct 2027**

Learn the Basics of Structured Query Language (SQL), Dicoding Indonesia — **Oct 2024 - Oct 2027**

Deep Learning, DeepLearning.AI — **Aug 2026**

DevOps Training, ID-Network — **Oct 2026**